Problem 12

The regions bounded by the curve $y = e^{-x}\sqrt{\sin x}$ $(x0)$ and the x-axis are rotated about the x-axis. Find the sum of the volumes of the solids generated by this rotation.

Solution

Step 1: Identifying the Intervals

The function $y = e^{-x}\sqrt{\sin x}$ is defined only when $\sin x \geq 0$. For $x \geq 0$, this occurs on the intervals:

$$[0, \pi], \quad [2\pi, 3\pi], \quad [4\pi, 5\pi], \quad \ldots$$

Generally, these are the intervals $[2k\pi, (2k+1)\pi]$ for $k = 0, 1, 2, \ldots$

On each of these intervals, the curve bounds a region with the x-axis, and rotation about the x-axis generates a solid whose volume must be calculated.

Step 2: Formula for Volume of Revolution

The volume of a solid generated by rotating the region bounded by the curve $y = f(x)$ and the x-axis about the x-axis on the interval $[a, b]$ is:

$$V = \pi \int_a^b y^2\, dx = \pi \int_a^b [f(x)]^2\, dx.$$

In our case:

$$y^2 = \left(e^{-x}\sqrt{\sin x}\right)^2 = e^{-2x}\sin x.$$

Step 3: Volume on a Single Interval

The volume of the solid on the interval $[2k\pi, (2k+1)\pi]$ is:

$$V_k = \pi \int_{2k\pi}^{(2k+1)\pi} e^{-2x}\sin x\, dx.$$

Step 4: Evaluating the Integral $\int e^{-2x}\sin x\, dx$

We use integration by parts twice. Let:

$$I = \int e^{-2x}\sin x\, dx.$$

First integration by parts:

$$u = \sin x, \quad dv = e^{-2x}dx \quad \Rightarrow \quad du = \cos x\, dx, \quad v = -\frac{1}{2}e^{-2x}.$$

$$I = -\frac{1}{2}e^{-2x}\sin x + \frac{1}{2}\int e^{-2x}\cos x\, dx.$$

Second integration by parts for $\int e^{-2x}\cos x\, dx$:

$$u = \cos x, \quad dv = e^{-2x}dx \quad \Rightarrow \quad du = -\sin x\, dx, \quad v = -\frac{1}{2}e^{-2x}.$$

$$\int e^{-2x}\cos x\, dx = -\frac{1}{2}e^{-2x}\cos x - \frac{1}{2}\int e^{-2x}\sin x\, dx = -\frac{1}{2}e^{-2x}\cos x - \frac{1}{2}I.$$

Substituting back:

$$I = -\frac{1}{2}e^{-2x}\sin x + \frac{1}{2}\left(-\frac{1}{2}e^{-2x}\cos x - \frac{1}{2}I\right).$$

$$I = -\frac{1}{2}e^{-2x}\sin x - \frac{1}{4}e^{-2x}\cos x - \frac{1}{4}I.$$

$$I + \frac{1}{4}I = -\frac{1}{2}e^{-2x}\sin x - \frac{1}{4}e^{-2x}\cos x.$$

$$\frac{5}{4}I = -\frac{1}{4}e^{-2x}(2\sin x + \cos x).$$

$$I = -\frac{1}{5}e^{-2x}(2\sin x + \cos x) + C.$$

Step 5: Definite Integral on the Interval $[2k\pi, (2k+1)\pi]$

$$\int_{2k\pi}^{(2k+1)\pi} e^{-2x}\sin x\,dx = \left[-\frac{1}{5}e^{-2x}(2\sin x + \cos x)\right]_{2k\pi}^{(2k+1)\pi}.$$

At the boundaries:

$$x = (2k+1)\pi: \quad \sin((2k+1)\pi) = 0, \quad \cos((2k+1)\pi) = -1$$
$$\Rightarrow \quad -\frac{1}{5}e^{-2(2k+1)\pi}(0-1) = \frac{1}{5}e^{-2(2k+1)\pi}.$$

$$x = 2k\pi: \quad \sin(2k\pi) = 0, \quad \cos(2k\pi) = 1$$
$$\Rightarrow \quad -\frac{1}{5}e^{-4k\pi}(0+1) = -\frac{1}{5}e^{-4k\pi}.$$

Therefore:

$$\int_{2k\pi}^{(2k+1)\pi} e^{-2x}\sin x\,dx = \frac{1}{5}e^{-2(2k+1)\pi} - \left(-\frac{1}{5}e^{-4k\pi}\right) = \frac{1}{5}e^{-4k\pi}(e^{-2\pi}+1).$$

Simplifying:

$$e^{-2(2k+1)\pi} + e^{-4k\pi} = e^{-4k\pi}e^{-2\pi} + e^{-4k\pi} = e^{-4k\pi}(e^{-2\pi}+1).$$

Step 6: Volume on a Single Interval

$$V_k = \pi\cdot\frac{1}{5}e^{-4k\pi}(1+e^{-2\pi}) = \frac{\pi(1+e^{-2\pi})}{5}e^{-4k\pi}.$$

Step 7: Total Volume

The sum of all volumes is:

$$V = \sum_{k=0}^{\infty} V_k = \sum_{k=0}^{\infty}\frac{\pi(1+e^{-2\pi})}{5}e^{-4k\pi} = \frac{\pi(1+e^{-2\pi})}{5}\sum_{k=0}^{\infty}\left(e^{-4\pi}\right)^k.$$

This is a geometric series with first term $a = 1$ and ratio $r = e^{-4\pi} < 1$:

$$\sum_{k=0}^{\infty}\left(e^{-4\pi}\right)^k = \frac{1}{1-e^{-4\pi}} = \frac{e^{4\pi}}{e^{4\pi}-1}.$$

Therefore:

$$V = \frac{\pi(1+e^{-2\pi})}{5}\cdot\frac{e^{4\pi}}{e^{4\pi}-1} = \frac{\pi e^{4\pi}(1+e^{-2\pi})}{5(e^{4\pi}-1)}.$$

Simplifying the numerator:

$$e^{4\pi}(1+e^{-2\pi}) = e^{4\pi} + e^{2\pi}.$$

Final Answer

The total volume of all solids of revolution is:

$$\boxed{V = \frac{\pi(e^{4\pi} + e^{2\pi})}{5(e^{4\pi} - 1)} = \frac{\pi e^{2\pi}(e^{2\pi} + 1)}{5(e^{4\pi} - 1)}}.$$

Alternatively, we can write:

$$\boxed{V = \frac{\pi(1 + e^{-2\pi})}{5(1 - e^{-4\pi})}}.$$